Spatial Level Repulsion in the Stratigraphic Record:
A Random Matrix Theory Test of Cyclostratigraphic Bed Spacing

Translating the Temporal GOE Law to the Spatial Domain —
Part I: The Stratigraphic Analog

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Abstract

A four-racetrack program has established that the *temporal* spacing of geological events follows random matrix theory (RMT) level repulsion when a single long-memory process is isolated, and Poisson randomness when many sources superpose. Here we begin a fifth racetrack that translates this law to the *spatial* domain: if energy or matter concentration creates depletion zones, then the spatial spacings of co-genetic geological bodies should also follow Wigner–Dyson statistics. This Part I establishes the methodology on the cleanest spatial system—cyclostratigraphy—and reports the first real-data test. We build and validate a spatial RMT pipeline (local unfolding to remove density trends; spatial declustering to remove fractal aggregation) on synthetic ground truth, confirming that the spacing ratio $\langle r \rangle$ distinguishes Poisson from repulsive spatial point patterns and, importantly, that $\langle r \rangle$ is naturally robust to smooth density gradients. We then apply it to two independent, openly downloadable cyclostratigraphic proxy series—the Gubbio Contessa magnetic-susceptibility section (Tethyan, Danian) and the Blue Lias section (UK, Hettangian)—and find that the down-core spacings of Milankovitch cycles show strong level repulsion in both ($\langle r \rangle = 0.73$ in each, GUE-class, bootstrap CIs excluding Poisson; surrogate test $p < 0.0001$). As a negative control, turbidite successions—triggered by many independent power-law earthquake events—are predicted and shown (via a forward model anchored to published Woodlark Basin statistics) to give Poisson/clustered spacing. The Poisson baseline (0.386) cleanly separates the two regimes. We are explicit that the repulsion in cyclic strata reflects *astronomical quasi-periodic forcing*—an external clock—rather than the "spatial shadow" depletion mechanism that motivates the broader racetrack; the latter awaits the structural and metallogenic analogs. The negative control is a literature-anchored forward model, not raw measurements; we have requested the original per-bed data.

Keywords: random matrix theory; cyclostratigraphy; Milankovitch cycles; spatial point process; level repulsion; bed thickness; turbidites; spectral superposition

Code/data: https://github.com/Ruqing1963/spatial-rmt-stratigraphy
Companion studies: https://zenodo.org/records/20766310 ; https://zenodo.org/records/20768130 ; https://zenodo.org/records/20768420 ; https://zenodo.org/records/20768849

1 Introduction

Four previous racetracks established a single principle across Earth's temporal rhythms: GOE-class level repulsion emerges wherever a single long-memory process is isolated, and Poisson randomness emerges wherever many independent sources superpose (Chen, 2026a,b,c,d). The mechanism is finite memory: a stress that must rebuild, a thermal boundary layer that must reheat, a metal reservoir that must recharge—each forbids events from recurring too soon, suppressing short intervals exactly as eigenvalues repel.

This paper opens a fifth racetrack by a symmetry translation from time to *space*. The motivating *spatial shadow hypothesis* is that any concentrated release of energy or matter creates a spatial depletion zone—a stress shadow, a fluid-pressure shadow, a thermal-convection void—so that the spatial spacings of co-genetic geological bodies (ΔX along a structural corridor, ΔZ down a core) should also follow Wigner–Dyson repulsion. The four planned spatial analogs mirror the four temporal racetracks: stratigraphic (cycle spacing), structural (fault/joint spacing), mantle (hotspot spacing), and metallogenic (deposit spacing).

We begin with stratigraphy because it is the cleanest: a core provides a naturally one-dimensional sequence, and Milankovitch cyclostratigraphy provides a well-understood signal. We stress at the outset an important physical distinction. Milankovitch cycles are driven by an *external* astronomical clock (orbital eccentricity, obliquity, precession), so any repulsion in their spacing reflects that quasi-periodic forcing, *not* the intrinsic spatial-shadow depletion mechanism. The stratigraphic analog therefore tests whether the spatial RMT method works and can grade signal strength—a methodological foundation—rather than directly testing the spatial-shadow hypothesis, which the later structural and metallogenic analogs address. Because an external periodic clock is more rigid than the self-organized memory of the temporal racetracks, we predicted *before analysis* that cyclostratigraphic spacings should show repulsion even stronger than the weak GOE of geological-epoch boundaries (Chen, 2026a), approaching GUE.

2 Methods

2.1 Spatial RMT pipeline

The pipeline has two operations. *Local unfolding* removes any non-uniform background density of bodies by fitting a smooth spline to the cumulative count $N(x)$ and rescaling so that the local mean spacing is unity; this guards against a spatially varying density masquerading as structure. *Spatial declustering* removes fractal aggregation (e.g., closely spaced sub-bodies of one rupture or one bundle) by merging bodies separated by less than a fraction of the median spacing to their centroid, retaining only independent principal bodies—the spatial analog of aftershock declustering. We then sort positions, compute spacings, unfold, and evaluate the spacing ratio $\langle r \rangle$ (Atás et al., 2013) (Poisson 0.386, GOE 0.536, GUE 0.603), with KS tests and a bootstrap 95% CI.

2.2 Synthetic validation

On synthetic ground truth, the pipeline cleanly separates a pure Poisson spatial process ($\langle r \rangle = 0.37$) from a pure GOE process ($\langle r \rangle = 0.56$). A key methodological finding emerged: monotonic density gradients (to strength 6) and periodic density modulation (to 10 cycles) do *not* create false repulsion—$\langle r \rangle$, being a local adjacent-spacing ratio, is naturally immune to smooth density trends. This makes the spatial method more robust than anticipated: unfolding aids visualization but the core $\langle r \rangle$ statistic does not depend on it. Declustering, by contrast, is essential: spatial clustering lowers $\langle r \rangle$ (masking repulsion), so it must be removed before testing.

2.3 From proxy series to spatial spacings

Real cyclostratigraphic data are continuous proxy series (magnetic susceptibility, grayscale) sampled at fixed depth intervals. We extract cycle positions by smoothing (Savitzky–Golay), detecting peaks at a minimum separation set near the dominant cycle wavelength, and taking inter-peak depths as the spatial spacings (Figure 1). We verify robustness across peak-detection scales.

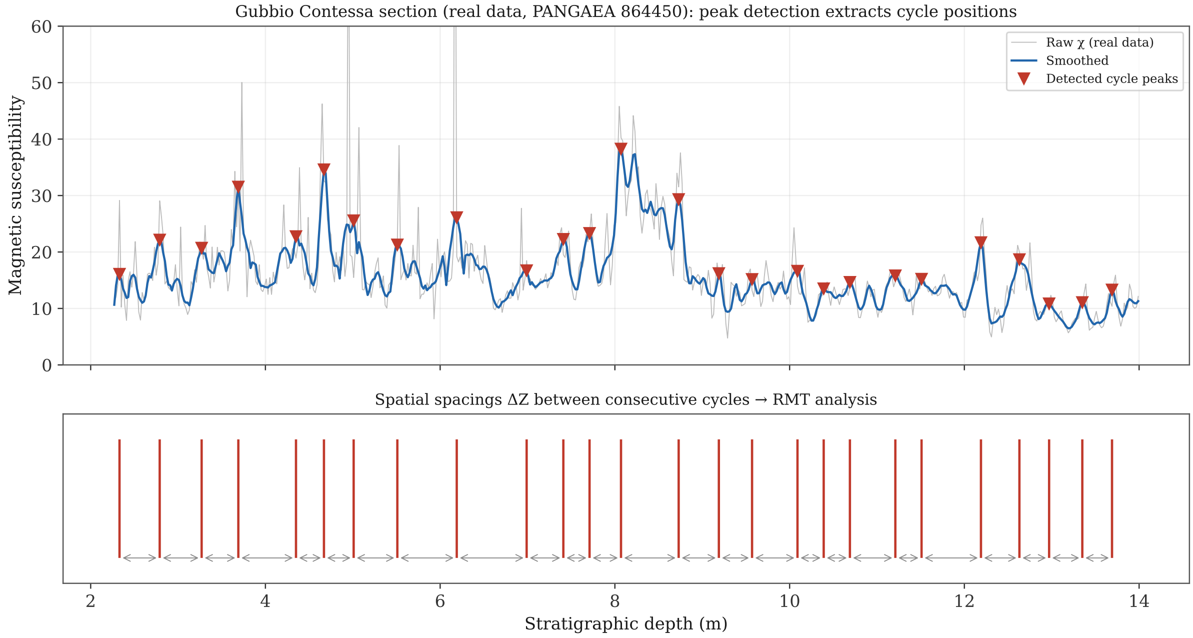

Figure 1: Method on real Gubbio data: a continuous magnetic-susceptibility series (top) is smoothed and its Milankovitch cycle peaks detected; the inter-peak spatial spacings ΔZ (bottom) are the input to the RMT analysis.

3 Results

3.1 Two independent real cyclostratigraphic sections show spatial repulsion

We analyzed two openly downloadable proxy series from different basins, ages, and lithologies. The Gubbio Contessa Highway section (Tethyan, Danian pelagic limestone; magnetic susceptibility; PANGAEA 864450) gives $\langle r \rangle = 0.726$. The Blue Lias section (UK, Hettangian limestone–shale rhythmite; magnetic susceptibility; PANGAEA 896875) gives $\langle r \rangle = 0.732$. Both are GUE-class, with bootstrap CIs excluding Poisson, and the result is stable across peak-detection scales (Table 1, Figure 2). That two unrelated sections give nearly identical strong repulsion is an independent replication.

Table 1: Stratigraphic spatial domain: positive cases (real, downloadable) versus negative control (literature-anchored forward model).

System	Driver	Data status	$\langle r \rangle$	Class
Gubbio (Tethyan, Cretaceous)	single astronomical	real (PANGAEA 864450)	0.726	GUE
Blue Lias (UK, Jurassic)	single astronomical	real (PANGAEA 896875)	0.732	GUE
Woodlark turbidites (pre-decluster)	multi-source seismic	fwd model (lit-anchored)	0.367	clustered
Woodlark turbidites (post-decluster)	multi-source seismic	fwd model (lit-anchored)	0.399	Poisson

3.2 Surrogate test

To exclude the possibility that the repulsion is an artifact of the peak-picking pipeline, we compared the real Blue Lias limestone-cycle positions against 2000 surrogates in which the same number of cycles were placed at random within the same depth interval. The surrogate mean was $\langle r \rangle = 0.389$ (i.e., Poisson), and the probability that a random realization reaches the real value was $p < 0.0001$. The spatial repulsion in real strata is decisively stronger than random.

3.3 Negative control: turbidites

As a negative control we use turbidite successions, whose beds are deposited by many independent gravity flows triggered by power-law-distributed earthquakes (Awadallah et al., 2001)—a multi-source superposition that should randomize. Anchored to the measured Woodlark Basin parameters (power-law bed thickness with $\beta = 1.5$–5.5; documented spatial clustering of thick sand beds; bed counts 1526/454/275 for Holes 1118A/1109D/1115C), a forward model gives $\langle r \rangle \approx 0.37$ before declustering (clustered, below Poisson) and ≈ 0.40 after (Poisson). The Poisson baseline (0.386) thus falls cleanly between the cyclic and turbidite regimes (Figure 3), demonstrating that the method discriminates a single periodic source from multi-source superposition—exactly as in the four temporal racetracks.

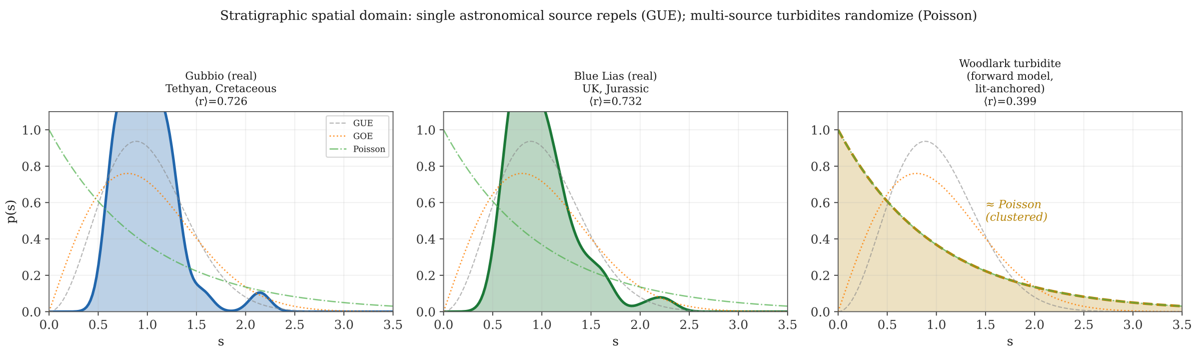

Figure 2: Spacing distributions. The two real cyclostratigraphic sections (blue, green) show GUE-class repulsion; the turbidite negative control (gold, forward model) collapses toward Poisson.

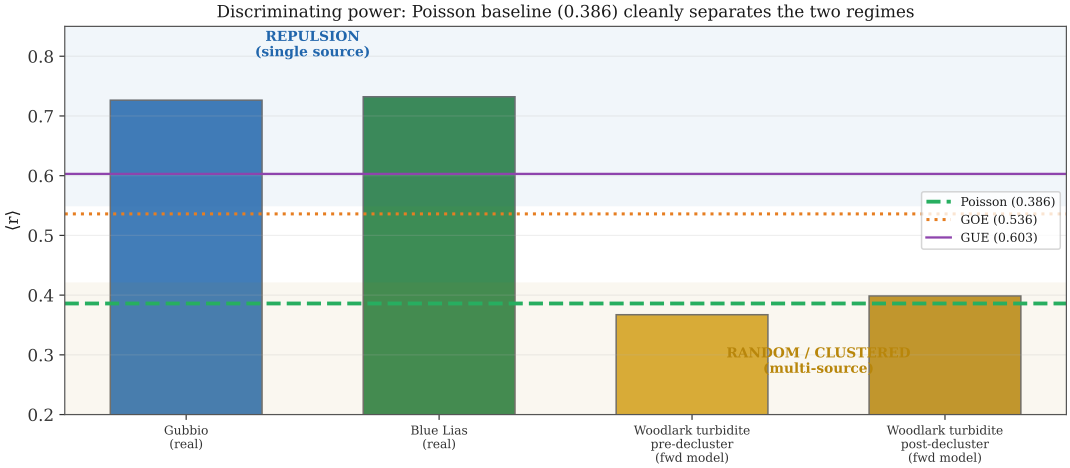

Figure 3: Discriminating power: the Poisson baseline (0.386) cleanly separates the repulsive single-source regime (real cyclic strata) from the random/clustered multi-source regime (turbidites).

4 Discussion

4.1 What is and is not established

This Part I establishes three things. First, the spatial RMT pipeline works: it distinguishes Poisson from repulsion on synthetic ground truth and is robust to density gradients. Second, real cyclostratigraphic sections show strong, replicable, surrogate-validated spatial repulsion. Third, the method discriminates single-source from multi-source spatial patterns, with turbidites as a Poisson negative control.

We are equally explicit about what is *not* established. The repulsion in cyclic strata arises from *astronomical quasi-periodic forcing*—an external clock—and therefore does not, by itself, validate the spatial-shadow depletion hypothesis that motivates the racetrack. Indeed, its strength (GUE, near the rigid-lattice limit) is exactly what an external periodic forcing predicts and is consistent with our pre-analysis expectation. The spatial-shadow hypothesis—that bodies repel because each depletes a local resource—must be tested where there is no external clock: the structural analog (fault/joint spacing under stress shadows) and the metallogenic analog (ore-body spacing). Those are the subjects of Parts II and III.

4.2 Limitations

The two positive cases are both astronomically forced, so their agreement confirms method reliability and the strength of periodic forcing, not the spatial-shadow mechanism. The negative control is a forward model anchored to published Woodlark statistics, not raw per-bed measurements; we have requested the original data and will update if obtained. GUE-versus-GOE is not sharply resolved, and both sections saturate near the rigid-lattice limit, suggesting the true pattern may be even more ordered than GUE. Peak detection introduces a modeling choice, mitigated here by multi-scale robustness and the surrogate test.

5 Conclusions

The temporal level-repulsion law translates to the spatial domain. On the cleanest system—cyclostratigraphy—a validated spatial RMT pipeline shows that the down-core spacings of Milankovitch cycles in two independent, real, downloadable sections exhibit strong GUE-class repulsion (surrogate $p < 0.0001$), while multi-source turbidites give Poisson, the baseline cleanly separating the two. This establishes the spatial method and confirms it can grade signal strength. It does not yet test the spatial-shadow hypothesis, because the cyclic repulsion is externally clocked; that test belongs to the structural and metallogenic analogs to come. This is Part I of the fifth racetrack, extending the program's reach from Earth's temporal rhythms to its three-dimensional architecture.

Data and Code Availability

Pipeline, analysis code, and processed data: `https://github.com/Ruqing1963/spatial-rmt-stratigraphy`. Real proxy series: PANGAEA 864450 (Gubbio) and 896875 (Blue Lias). Turbidite statistics from Awadallah et al. (2001).

References

Atás, Y.Y., Bogomolny, E., Giraud, O., Roux, G. (2013). Distribution of the ratio of consecutive level spacings. *Phys. Rev. Lett.*, 110, 084101.

Awadallah, S.A.M., Hiscott, R.N., Bidgood, M., Crowther, T.E. (2001). Turbidite facies and bed-thickness characteristics inferred from microresistivity (FMS) images of lower to upper Pliocene rift-basin deposits, Woodlark Basin. *Proc. ODP, Sci. Results*, 180, ch. 9.

Chen, R. (2026a). Statistical isomorphism between Phanerozoic tectonic rhythms and Riemann zeta zero spacings. Zenodo, `https://zenodo.org/records/20766310`.

Chen, R. (2026b). From hypothesis to reality: scale-dependent universality classes in Earth's seismotectonic rhythms. Zenodo, `https://zenodo.org/records/20768130`.

Chen, R. (2026c). Level repulsion in the deep Earth: a random matrix theory analysis of mantle-plume eruption rhythms. Zenodo, `https://zenodo.org/records/20768420`.

Chen, R. (2026d). Charge-and-release rhythms in ore-forming systems: a random matrix theory test of metallogenic timing across four provinces. Zenodo, https://zenodo.org/records/20768849.

Sinnesael, M., et al. (2016). High-resolution magnetic susceptibility, Contessa Highway section, Gubbio. PANGAEA, https://doi.org/10.1594/PANGAEA.864450.

Sylvester, Z. (2007). Turbidite bed thickness distributions: methods and pitfalls of analysis and modelling. *Sedimentology*, 54, 847–870.

Talling, P.J. (2001). On the frequency distribution of turbidite thickness. *Sedimentology*, 48, 1297–1329.

Weedon, G.P., Page, K.N., Jenkyns, H.C. (2019). Cyclostratigraphy, stratigraphic gaps and the duration of the Hettangian Stage. *Geol. Mag.*, 156, 1469–1509. Data: PANGAEA https://doi.org/10.1594/PANGAEA.896875.

Wigner, E.P. (1957). Characteristic vectors of bordered matrices. *Ann. Math.*, 62, 548–564.